

THE ORDER OF A 5-CHROMATIC UNIT-DISTANCE GRAPH IS AT LEAST 29

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ABSTRACT. Let v_5 be the least number of vertices of a 5-chromatic unit-distance graph in the Euclidean plane. The best published bound is $v_5 \geq 28$, due to de Grey and Parts. We observe that their own edge bound, combined with the unit-distance counts of Alexeev, Mixon and Parshall published two years later, gives $v_5 \geq 29$ with no new computation. The argument is two lines; what it needs is a table that did not exist when the vertex bound was derived. We also record exactly what 30 and 31 would require, and note that one of the two obstacles de Grey and Parts name for their own method is removable for polygonal tilings.

1. THE TWO INGREDIENTS

Write $u(n)$ for the maximum number of unit distances determined by n points in the plane, and e_5 for the least number of edges of a 5-chromatic unit-distance graph in $\mathbb{R}^2$.

Theorem 1 (de Grey and Parts [2]). $e_5 \geq 99$ and $v_5 \geq 28$.

Their bound on e_5 comes from a refinement of Pritikin’s uncovered-density scheme together with the Gwyn–Stavrianos badness argument: if p_4 is the probability that a random unit segment is monochromatic under the best 4-colouring of the plane available to the method, then $e_5 \geq \lceil 1/p_4 \rceil$. They state the two bounds together, in the form “we get the bounds: $e_5 \geq 99$, $v_5 \geq 28$, $e_6 \geq 182$, $v_6 \geq 42$ ”.

Theorem 2 (Alexeev, Mixon and Parshall [1]). For $22 \leq n \leq 30$,

n	22	23	24	25	26	27	28	29	30
$u(n) \geq$	60	64	68	72	76	81	85	89	93
$u(n) \leq$	61	66	72	78	84	90	96	103	110

The lower bounds are earlier constructions; the upper bounds are that paper’s contribution.

2020 *Mathematics Subject Classification.* 05C15, 52C10, 05C62.

Key words and phrases. Hadwiger–Nelson problem, unit-distance graph, chromatic number of the plane, Erdős unit distance problem.

2. THE OBSERVATION

Proposition 3. $v_5 \geq 29$.

Proof. Suppose G is a 5-chromatic unit-distance graph in the plane on $n \leq 28$ vertices. Its edges are unit distances among n points, so $|E(G)| \leq u(n)$. The function u is non-decreasing, since adding a point removes no unit distance, so $|E(G)| \leq u(28) \leq 96$ by Theorem 2. But $|E(G)| \geq e_5 \geq 99$ by Theorem 1, and $96 < 99$. $\square$

Remark 4. The bound is exactly 29 by this route and no further: $u(29) \leq 103$, which does not contradict $e_5 \geq 99$.

3. WHY IT WAS AVAILABLE BUT NOT TAKEN

De Grey and Parts convert edges to vertices using the bounds of Ágoston and Pálvölgyi, which in the relevant range allow at most 97 edges on 27 vertices and at most 102 on 28. So 27 was excluded and 28 was not: their bound is exactly what their inputs allow. Their paper is from March 2023; [1] is from December 2024, revised February 2025, and is strictly better at every n in this range. At $n = 28$ it replaces 102 by 96, and 28 no longer survives.

Nothing else is needed, and nothing else is claimed. This is an observation about two existing results, not a new method.

4. WHAT 30 AND 31 REQUIRE

Two independent levers, both engineering rather than new theory.

Lever 1: raise e_5 . Against the table of Theorem 2, held fixed,

$$e_5 \geq 104 \implies u(29) \leq 103 < 104 \implies v_5 \geq 30,$$

$$e_5 \geq 111 \implies u(30) \leq 110 < 111 \implies v_5 \geq 31.$$

Reaching $e_5 \geq 104$ needs $p_4 < 1/104 = 0.0096154$, about 5% below the value behind 99. De Grey and Parts name their own obstacle: they compute p_4 by numerical integration and write that they “encountered some problems with the numerical integration function `NIntegrate` in the general case, and were forced to limit ourselves to relatively simple tilings”, nominating tilings with wavy borders as the promising family.

For a *polygonal* tiling that obstacle is removable. The integrand $M_k(\sigma, \varphi)$ of

$$p_k = \frac{1}{2\pi S} \int_S d\sigma \int_0^{2\pi} d\varphi M_k(\sigma, \varphi)$$

is piecewise constant: a unit segment’s two endpoints each lie in some cell, and M_k depends only on which pair of cells. So p_k is an exact finite sum of cell measures, computable by a planar arrangement rather than by quadrature. That makes richer tilings reachable and the answer exact rather than approximate.

Lever 2: tighten $u(29)$ and $u(30)$. No colouring work at all. Theorem 2's own lower bounds leave room: $u(29)$ is known only to lie in $[89, 103]$ and $u(30)$ in $[93, 110]$. Either $u(29) \leq 98$ or $u(30) \leq 98$ gives $v_5 \geq 30$ or $v_5 \geq 31$ at once, and both targets sit inside the intervals that are currently open.

REFERENCES

- [1] B. Alexeev, D. G. Mixon and H. Parshall, *The Erdős unit distance problem for small point sets*, arXiv:2412.11914 (2024, revised 2025).
- [2] A. D. N. J. de Grey and J. Parts, *On lower bounds of the order of k -chromatic unit distance graphs*, Geombinatorics **32** (2022), no. 2; arXiv:2303.14714.

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